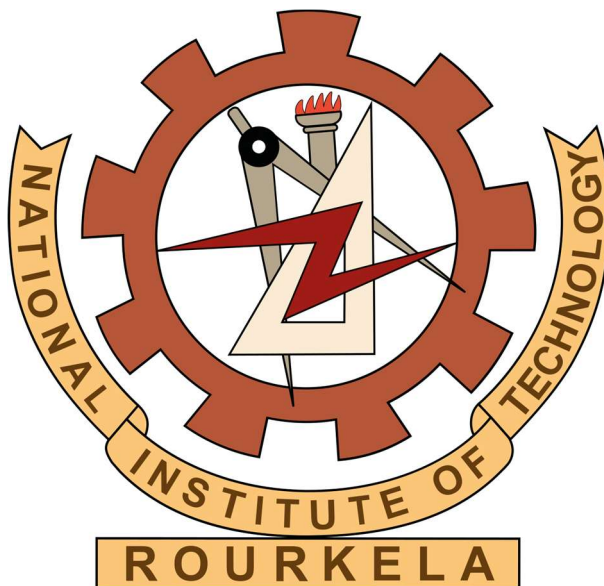




CHEMICAL ENGINEERING PROJECT REPORT

ISOTHERMAL EQUILIBRIUM REACTOR SIMULATION FOR ETHANOL PRODUCTION USING DWSIM

A steady-state equilibrium-reactor study of the reversible gas-phase hydration of ethylene

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Software: DWSIM

Primary simulation file: E-REACTOR.dwxmz

Simulation build recorded in file: DWSIM 10.2.4.0

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Note: Page numbers are intentionally omitted from this generated editable report so that they can update automatically after final institutional formatting.

1. Abstract

This project presents a steady-state simulation of ethanol formation by reversible gas-phase hydration of ethylene in an isothermal equilibrium reactor. The modeled reaction is $\text{C}_2\text{H}_4(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{C}_2\text{H}_5\text{OH}(\text{g})$. The feed consists of 50 mol/s ethylene, 50 mol/s water and no ethanol, giving a total feed rate of 100 mol/s. The reactor is operated at 145 °C (418.15 K) and 1 bar, with the reaction phase specified as vapor and an equilibrium constant of 0.1443. The DWSIM model uses the Raoult's Law property package recorded in the simulation file and an equilibrium reaction based on fugacity. At the specified conditions, the DWSIM outlet vapor flow is 96.806504 mol/s, containing 46.806504 mol/s ethylene, 46.806504 mol/s water and 3.193496 mol/s ethanol. The corresponding ethylene conversion is 6.38699%, reported by the reactor at approximately 6.39%. The handwritten calculation uses an approximate extent of reaction of 3.25 mol/s, giving a theoretical conversion of approximately 6.50%. Solving the same displayed equilibrium equation more precisely gives an extent of 3.25878 mol/s and a conversion of 6.51756%. The DWSIM value therefore agrees closely with the hand calculation, with a percentage difference of about 2.00% relative to the precise theoretical result. The study demonstrates how stoichiometry, equilibrium relationships and process simulation can be combined to quantify conversion while recognizing that equilibrium limits the extent of reaction.

2. Introduction

Ethanol is an important chemical product and the reaction studied here provides a direct route from ethylene and water. The hydration reaction is reversible, so the final composition is governed by chemical equilibrium rather than by assuming complete conversion of the limiting reactant. For a reversible reaction, the equilibrium condition links composition and thermodynamic driving force through an equilibrium constant. An equilibrium-reactor model is therefore appropriate when the objective is to determine the composition corresponding to the specified equilibrium condition.

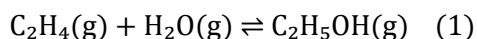
Process simulation is useful because it allows material flows, thermodynamic calculations, reaction specifications and energy requirements to be evaluated within one consistent flowsheet. DWSIM was selected because the supplied project itself was constructed in DWSIM and the saved simulation contains the reactor configuration, reaction definition, property-package selection and calculated stream results.

3. Objectives

- Model ethylene hydration to ethanol in DWSIM.
- Configure and evaluate an isothermal equilibrium reactor.
- Determine the equilibrium outlet composition for the specified feed.
- Calculate ethylene conversion from the outlet flow.
- Compare the theoretical extent-of-reaction calculation with the DWSIM result.
- Verify the component and total material balance around the reactor.
- Interpret the influence of equilibrium, pressure and temperature qualitatively without introducing unsupported sensitivity results.

4. Chemical Reaction and Stoichiometry

The reversible reaction represented in the supplied calculation sheet and DWSIM file is:



The stoichiometric coefficients are:

Component	Stoichiometric coefficient, v_i
C ₂ H ₄	-1
H ₂ O	-1
C ₂ H ₅ OH	+1

The change in total number of moles per mole of reaction is:

$$\Delta v = \sum v_i = (-1) + (-1) + 1 = -1 \quad (2)$$

Δv = change in total gas-phase moles per stoichiometric extent.

The negative value of Δv means that the reaction reduces the total number of gas-phase moles as it proceeds in the forward direction. Thus, at equilibrium, pressure can influence the composition because the stoichiometric reaction changes the number of gas-phase molecules.

5. Feed Conditions

Component	Feed flow rate (mol/s)	Feed mole fraction
C ₂ H ₄	50	0.5
H ₂ O	50	0.5
C ₂ H ₅ OH	0	0
Total	100	1.0

The feed mole fraction is obtained from:

$$y_i = n_i / n_0 \quad (3)$$

y_i = feed mole fraction of component i ; n_i = component molar flow rate; n_0 = total feed molar flow rate.

For ethylene, $y_{C_2H_4} = 50/100 = 0.5$. The same calculation gives $y_{H_2O} = 0.5$, while the ethanol feed mole fraction is zero.

6. Theoretical Equilibrium Calculation

Let ε denote the extent of reaction in mol/s. Applying the stoichiometric coefficients gives:

Component	Initial (mol/s)	Change (mol/s)	Equilibrium (mol/s)
C ₂ H ₄	50	$-\varepsilon$	$50 - \varepsilon$
H ₂ O	50	$-\varepsilon$	$50 - \varepsilon$
C ₂ H ₅ OH	0	$+\varepsilon$	ε
Total	100	$-\varepsilon$	$100 - \varepsilon$

Therefore, the equilibrium total molar flow is:

$$n = 100 - \varepsilon \quad (4)$$

n = total equilibrium molar flow rate.

The corresponding equilibrium mole fractions are:

$$y_{C_2H_4} = (50 - \varepsilon)/(100 - \varepsilon) \quad (5)$$

$y_{C_2H_4}$ = equilibrium mole fraction of ethylene.

$$y_{H_2O} = (50 - \varepsilon)/(100 - \varepsilon) \quad (6)$$

y_{H_2O} = equilibrium mole fraction of water.

$$y_{C_2H_5OH} = \varepsilon/(100 - \varepsilon) \quad (7)$$

$y_{C_2H_5OH}$ = equilibrium mole fraction of ethanol.

7. Equilibrium Constant Equation

For the gas-phase reaction, the supplied handwritten calculation expresses equilibrium in terms of activities. The general expression is:

$$K = a_{C_2H_5OH} / (a_{C_2H_4} a_{H_2O}) \quad (8)$$

K = equilibrium constant; a_i = activity of component i .

For the ideal-gas activity representation used in the hand calculation:

$$a_i = y_i(P/P^\circ) \quad (9)$$

P = system pressure; P° = standard pressure; y_i = gas-phase mole fraction.

At $P = 1$ bar and $P^\circ = 1$ bar, the pressure ratio is unity. The hand-calculation expression therefore reduces to the mole-fraction form:

$$K = y_{C_2H_5OH} / (y_{C_2H_4} y_{H_2O}) \quad (10)$$

Substitution of the extent expressions gives:

$$[\varepsilon/(100-\varepsilon)] / \{[(50-\varepsilon)/(100-\varepsilon)] [(50-\varepsilon)/(100-\varepsilon)]\} = 0.1443 \quad (11)$$

$$\varepsilon(100-\varepsilon)/(50-\varepsilon)^2 = 0.1443 \quad (12)$$

The handwritten sheet records an approximate solution $\varepsilon \approx 3.25$ mol/s. Solving the same equation numerically gives the physically meaningful root $\varepsilon = 3.258778754$ mol/s. The second mathematical root is approximately 96.7412 mol/s, which is not physically consistent with the stoichiometric availability of 50 mol/s ethylene and 50 mol/s water because ε cannot exceed 50 mol/s.

8. Ethylene Conversion

$$X_{C_2H_4} = [(n_{C_2H_4,0} - n_{C_2H_4,out})/n_{C_2H_4,0}] \times 100 \quad (13)$$

$X_{C_2H_4}$ = ethylene conversion (%); $n_{C_2H_4,0}$ = inlet ethylene flow; $n_{C_2H_4,out}$ = outlet ethylene flow.

$$X_{C_2H_4,theory} = (\varepsilon/50) \times 100 \quad (14)$$

Using the handwritten $\varepsilon = 3.25$ mol/s gives $X \approx 6.50\%$. Using the more precise root $\varepsilon = 3.258778754$ mol/s gives $X = 6.51756\%$. Both values are theoretical results; the first is the value explicitly recorded on the handwritten sheet.

9. DWSIM Simulation Methodology

The supplied DWSIM file records the following workflow and configuration:

1. Create a steady-state simulation and define the components Ethylene, Water and Ethanol.
2. Define the reaction as $CH_2CH_2 + HOH \rightleftharpoons CH_3CH_2OH$ and assign stoichiometric coefficients -1 , -1 and $+1$.
3. Define the reaction as an equilibrium reaction with reaction phase = Vapor and constant equilibrium value $K = 0.1443$.
4. Use the reaction set named ETHANOL (ID DefaultSet).
5. Add an Equilibrium Reactor and connect FEED, VAPOUR, LIQUID and the energy stream E1.
6. Operate the reactor in Isothermic mode with outlet temperature 418.15 K (145 °C).
7. Set pressure drop to 0 bar; the supplied stream data record the operating pressure as 100000 Pa (1 bar).

8. Use the Raoult's Law property package recorded in the simulation.
9. Run the steady-state calculation and read the outlet flow, composition and conversion.

The DWSIM XML records BuildVersion 10.2.4.0. The equilibrium reaction is stored with ReactionBasis = Fugacity, while the selected Raoult's Law property package records VaporPhaseFugacityCalculationMode = Ideal. These are properties of the supplied simulation file and are reported here without replacing them with another thermodynamic model.

E-REACTOR (Equilibrium Reactor)

Linked to

Connections

Inlet Stream	FEED		
Outlet Stream 1	VAPOUR		
Outlet Stream 2	LIQUID		
Energy Stream	E1		

Calculation Parameters

Parameters **Convergence**

Reaction Set	ETHANOL
Calculation Mode	Isothermic
Outlet Temperature	145 C
Pressure Drop	0 bar
Property Package	Raoult's Law (1)
External Solver	

Results

General **Reactions** **Conversions**

Compound	Conversion (%)
Ethylene	6.38699
Water	6.38699

Figure 1. DWSIM equilibrium-reactor configuration and calculated ethylene conversion.

10. DWSIM Flowsheet

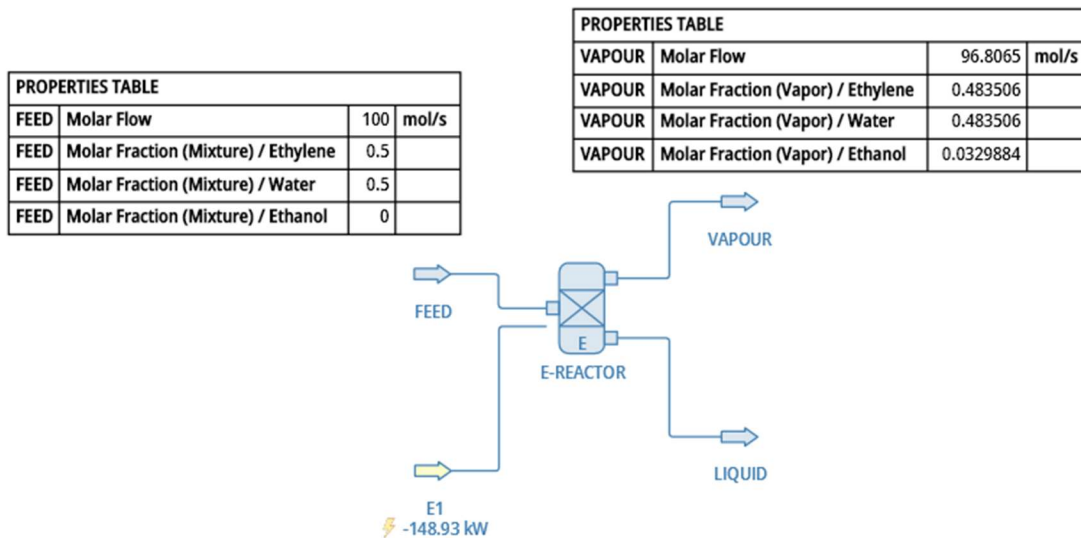


Figure 2. DWSIM flowsheet for the isothermal equilibrium reactor.

The supplied flowsheet contains:

Element	Role in the simulation
FEED	Material stream entering the equilibrium reactor.
E-REACTOR	Equilibrium reactor performing the specified reversible reaction.
VAPOUR	Calculated material outlet containing the reported vapor-phase product mixture.
LIQUID	Second reactor outlet; the supplied simulation records zero flow in this stream.
E1	Energy stream connected to the reactor; the file records an energy flow of -148.925571 kW.

The flowsheet therefore represents FEED → E-REACTOR → VAPOUR, with LIQUID as the second material outlet and E1 as the energy connection. The graphical property tables in the supplied screenshot show a 100 mol/s feed and a vapor outlet flow of approximately 96.8065 mol/s.

11. DWSIM Results

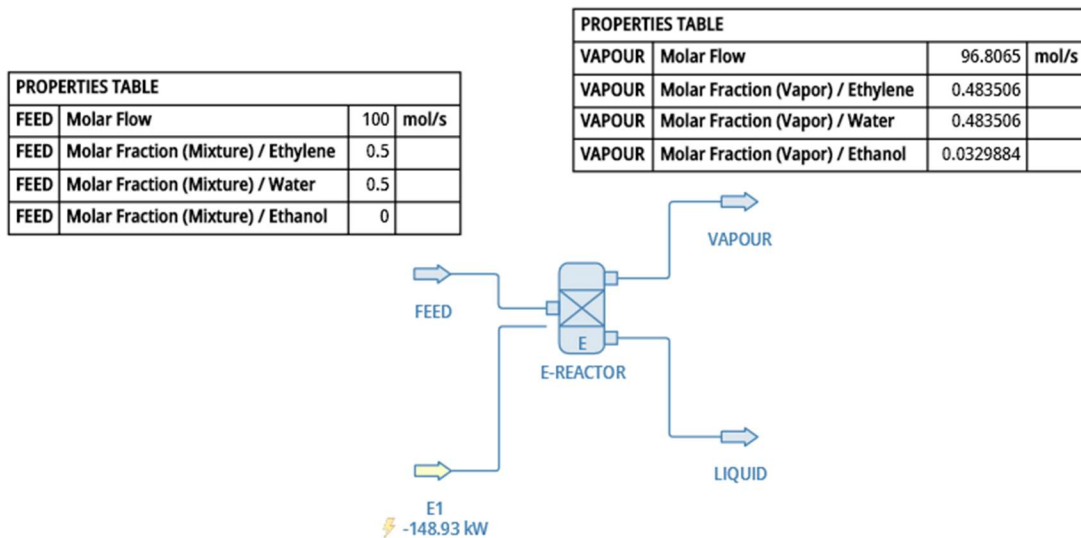


Figure 3. Supplied DWSIM flowsheet with feed and vapor property tables.

The DWSIM simulation file gives the following outlet component values:

Component	Outlet molar flow (mol/s)	Outlet mole fraction
C ₂ H ₄	46.806504172936	0.483505778592
H ₂ O	46.806504172936	0.483505778592
C ₂ H ₅ OH	3.193495827064	0.032988442815
Total	96.806504172936	1.000000000000

The component molar flows are obtained directly from the simulation stream data. For the displayed property-table values, the calculation is:

$$n_i = y_i n \quad (15)$$

n_i = component molar flow; y_i = outlet mole fraction; n = total vapor molar flow.

Ethylene: $n_{C_2H_4} = (0.483505778592)(96.806504172936) \approx 46.806504 \text{ mol/s}$.

Water: $n_{H_2O} = (0.483505778592)(96.806504172936) \approx 46.806504 \text{ mol/s}$.

Ethanol: $n_{C_2H_5OH} = (0.032988442815)(96.806504172936) \approx 3.193496 \text{ mol/s}$.

The internally stored values are used for the calculations above, avoiding unnecessary intermediate rounding. The reported vapor total is 96.806504 mol/s, and the liquid outlet has zero molar flow in the supplied simulation.

12. DWSIM Ethylene Conversion

$$X_{\text{DWSIM}} = [(nC_{2}H_{4,0} - nC_{2}H_{4,\text{out}})/nC_{2}H_{4,0}] \times 100 \quad (16)$$

Substituting the DWSIM outlet flow and the inlet ethylene flow:

$$X_{\text{DWSIM}} = [(50.000000 - 46.806504172936) / 50.000000] \times 100 = 6.38699165\% \quad (17)$$

The DWSIM reactor conversion panel reports 6.38699%, which is consistent with the calculation above.

13. Comparison Between Hand Calculation and DWSIM

Parameter	Hand calculation	DWSIM
Temperature	145 °C	145 °C
Pressure	1 bar	1 bar
Ethylene feed	50 mol/s	50 mol/s
Water feed	50 mol/s	50 mol/s
Equilibrium constant	0.1443	0.1443
Extent of reaction	≈3.25 mol/s	3.193495827 mol/s from outlet
Ethylene conversion	≈6.50%	6.38699%

$$\text{Percentage difference} = |DWSIM - \text{theoretical}| / \text{theoretical} \times 100 \quad (18)$$

Using the precise solution of the displayed equilibrium equation (6.51756%) as the theoretical reference:

$$\text{Percentage difference} = |6.38699 - 6.51756| / 6.51756 \times 100 = 2.00\% \quad (19)$$

The difference is small. Possible sources within the modeling framework include numerical solution details, finite numerical precision, the use of fugacity as the DWSIM reaction basis, the property-package implementation, and the fact that the handwritten calculation rounds the extent to 3.25 mol/s. The supplied file specifically records an ideal vapor-phase fugacity calculation mode within the selected Raoult's Law package; no different model is assumed here.

14. Material Balance

For each component at steady state:

$$\text{Input} + \text{Generation} = \text{Output} \quad (20)$$

Accumulation = 0 at steady state.

For the reaction $C_2H_4 + H_2O \rightleftharpoons C_2H_5OH$, one mole of ethylene and one mole of water are consumed per mole of ethanol formed. The DWSIM outlet data show equal consumption of ethylene and water:

Component	Inlet (mol/s)	Outlet (mol/s)	Change (mol/s)
C ₂ H ₄	50.000000	46.806504	-3.193496
H ₂ O	50.000000	46.806504	-3.193496
C ₂ H ₅ OH	0.000000	3.193496	3.193496

The reaction extent calculated from ethanol formation is 3.193495827064 mol/s. Therefore, the predicted ethylene and water outlet flows are each $50 - 3.193495827064 = 46.806504172936$ mol/s, which matches the DWSIM stream values.

Total molar flow check: inlet = 100.000000 mol/s; outlet = 96.806504172936 mol/s. The component sum agrees with the DWSIM total of 96.806504172936 mol/s.

Mass balance check: the file records an inlet mass flow of 2.303422 kg/s and an outlet mass flow of 2.303422 kg/s. The equality confirms the component mass balance within the stored numerical precision.

Energy information recorded in the simulation

The DWSIM file records reactor $\Delta Q = -148.925571055$ kW and the connected E1 energy stream carries the same numerical energy flow. The reaction definition also stores a reaction heat value of -45.646 kJ/mol. These are reported as simulation-file values; no additional interpretation of sign or reaction thermicity is imposed.

15. Equilibrium Reactor Theory

An equilibrium reactor differs from a conversion reactor in the way the reaction extent is specified. In a conversion reactor, conversion is normally an input or imposed design specification. In an equilibrium reactor, the outlet composition is obtained from the equilibrium condition together with material and thermodynamic relationships.

Chemical equilibrium corresponds to a state in which the forward and reverse reaction tendencies balance. For the reversible hydration reaction, this means that the composition does not continue toward complete ethanol formation at the specified conditions; instead, the reaction stops at an equilibrium extent consistent with the specified equilibrium constant.

The equilibrium constant provides the composition relationship. In the hand calculation, the relationship is expressed using activities and, at 1 bar with the stated standard pressure, reduces to a mole-fraction expression. In the DWSIM file, the reaction is stored with a fugacity basis, and the property package specifies ideal vapor-phase fugacity calculation mode.

No reaction-rate constants, residence time, catalyst model or kinetic expression are introduced because these are not required by the supplied equilibrium model and no supported kinetic data are present in the project material.

16. Effect of Pressure

For the reaction, $\Delta v = -1$. Increasing pressure generally favors the side with fewer gas-phase moles. Because the product side contains one mole of gas compared with two moles on the reactant side, the equilibrium tendency is qualitatively toward ethanol when pressure is increased, all else being equal.

No numerical pressure-sensitivity study is included because no such sweep is present in the supplied simulation. Therefore, only the qualitative equilibrium interpretation is reported.

17. Effect of Temperature

Temperature affects the equilibrium constant and therefore the equilibrium composition. The present project fixes the reactor at 145 °C (418.15 K) and uses the specified constant $K = 0.1443$. Without introducing external thermodynamic data or an additional temperature-sensitivity study, the direction of the temperature effect on conversion is not assigned here.

18. Limitations

- The model assumes chemical equilibrium rather than explicitly predicting conversion from reaction kinetics.
- The flowsheet contains a single equilibrium reactor representation and does not model reactor residence-time distribution.

- No explicit catalyst model is included in the supplied project.
- The result depends on the thermodynamic/property-package formulation stored in the DWSIM file.
- The selected reaction is represented with a constant equilibrium value of 0.1443 in the supplied model; temperature-dependent K calculations are not introduced here.
- The simulation does not include downstream ethanol purification or separation design beyond the reactor's vapor/liquid outlet representation.
- No economic analysis, plant-scale sizing or industrial performance claim is made.
- No experimental validation is included.
- All reported results correspond specifically to the supplied feed and reactor conditions.

19. Conclusion

The supplied DWSIM project successfully represents the reversible hydration of ethylene to ethanol in an isothermal equilibrium reactor. The feed contains 50 mol/s ethylene and 50 mol/s water, and the reactor operates at 145 °C and 1 bar with $K = 0.1443$. The DWSIM equilibrium calculation produces 46.806504 mol/s ethylene, 46.806504 mol/s water and 3.193496 mol/s ethanol in the vapor outlet, for a total vapor flow of 96.806504 mol/s.

The resulting ethylene conversion is 6.38699%. The handwritten calculation gives approximately 3.25 mol/s extent and 6.50% conversion. Solving the same displayed equation without the handwritten rounding gives 3.258779 mol/s and 6.51756% conversion, so the DWSIM result differs by approximately 2.00% relative to the precise theoretical value.

The close agreement demonstrates consistency between the stoichiometric equilibrium calculation and the DWSIM simulation. The study also highlights the importance of equilibrium limitations: the reversible reaction does not convert the entire ethylene feed to ethanol at the specified equilibrium condition. The conclusions are limited to the supplied simulation and should not be interpreted as experimental or industrial-scale validation.

20. References

[1] DWSIM simulation file supplied for this project: E-REACTOR.dwxmz. The saved file records DWSIM BuildVersion 10.2.4.0, the equilibrium reactor configuration, reaction definition, property package and calculated stream results.

[2] Uploaded handwritten calculation sheet: "Isothermal Equilibrium Reactor (Ethanol Production)", containing the reaction stoichiometry, extent-of-reaction calculation, equilibrium relation and approximate conversion.

[3] Uploaded DWSIM reactor-configuration screenshot showing the Equilibrium Reactor settings, isothermal operation at 145 °C, pressure drop of 0 bar, Raoult's Law (1) property package and calculated ethylene conversion.

[4] Uploaded DWSIM flowsheet/result screenshot showing FEED, E-REACTOR, VAPOUR, LIQUID and E1 together with feed and vapor property tables.

[5] Uploaded handwritten DWSIM methodology sheet showing the sequence used to configure the equilibrium reactor and obtain the ethylene conversion.

Note: No external journal articles, DOIs, page numbers or unsupported thermodynamic references have been introduced. The report is intentionally grounded in the supplied simulation file and project images/notes.

21. Appendix

Appendix A. Full theoretical calculation

Initial feed:

$$n_{\text{C}_2\text{H}_4,0} = 50 \text{ mol/s}; n_{\text{H}_2\text{O},0} = 50 \text{ mol/s}; n_{\text{C}_2\text{H}_5\text{OH},0} = 0 \text{ mol/s}; n_0 = 100 \text{ mol/s}.$$

Equilibrium:

$$n_{\text{C}_2\text{H}_4} = 50 - \varepsilon; n_{\text{H}_2\text{O}} = 50 - \varepsilon; n_{\text{C}_2\text{H}_5\text{OH}} = \varepsilon; n = 100 - \varepsilon.$$

At $P = P^\circ = 1 \text{ bar}$:

$$K = [\varepsilon/(100-\varepsilon)] / \{[(50-\varepsilon)/(100-\varepsilon)]^2\} = \varepsilon(100-\varepsilon)/(50-\varepsilon)^2 = 0.1443.$$

Physical solution: $\varepsilon = 3.258778754470 \text{ mol/s}$; $X = 6.51755751\%$.

Appendix B. DWSIM reactor configuration

Item	Recorded value
Reactor type	Equilibrium Reactor
Operation	Isothermic
Outlet temperature	418.15 K (145 °C)
Pressure drop	0 bar
Operating pressure in stored stream data	100000 Pa (1 bar)
Reaction set	ETHANOL / DefaultSet
Reaction basis	Fugacity
Reaction phase	Vapor
K	0.1443
Property package	Raoult's Law (1)
Vapor fugacity calculation mode	Ideal
Reactor solver setting	Use IPOPT = true
E1 energy flow	-148.925571 kW

Appendix C. DWSIM Result Screenshots

E-REACTOR (Equilibrium Reactor)

Linked to

Connections

Inlet Stream	FEED		
Outlet Stream 1	VAPOUR		
Outlet Stream 2	LIQUID		
Energy Stream	E1		

Calculation Parameters

Parameters **Convergence**

Reaction Set	ETHANOL
Calculation Mode	Isothermic
Outlet Temperature	145 C
Pressure Drop	0 bar
Property Package	Raoult's Law (1)
External Solver	

Results

General **Reactions** **Conversions**

Compound	Conversion (%)
Ethylene	6.38699
Water	6.38699

Figure 4. DWSIM Equilibrium Reactor configuration and conversion result.

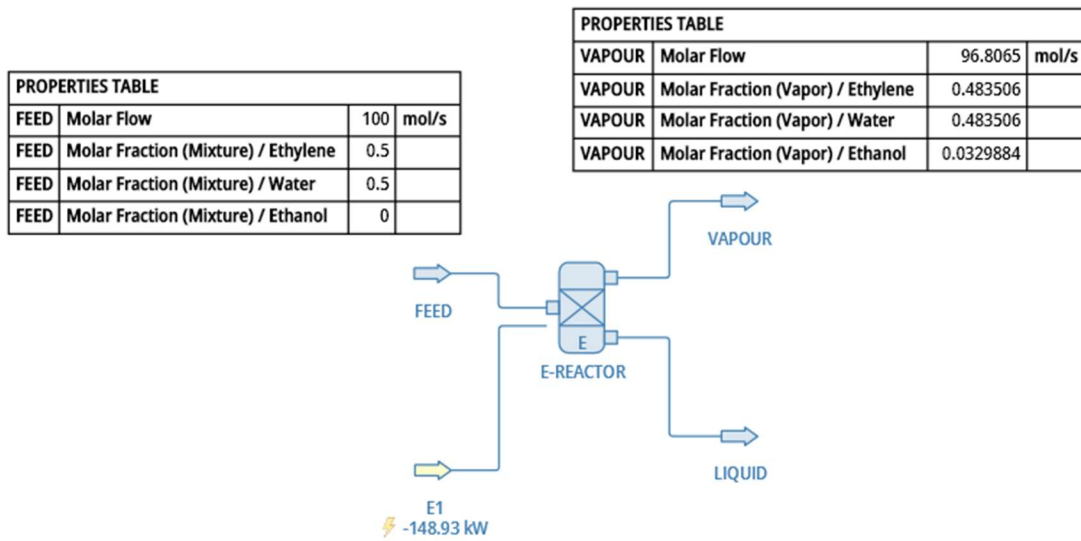


Figure 5. DWSIM flowsheet and property tables.

Important project inputs consolidated from the supplied material:

Input	Value
Reaction	$C_2H_4(g) + H_2O(g) \rightleftharpoons C_2H_5OH(g)$
Ethylene feed	50 mol/s
Water feed	50 mol/s
Ethanol feed	0 mol/s
Total feed	100 mol/s
Temperature	145 °C = 418.15 K
Pressure	1 bar
Equilibrium constant	0.1443
Reaction phase	Vapor
Reactor operation	Isothermic
Outlet vapor flow	96.806504 mol/s
Ethylene conversion	6.38699%

End of report.

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